

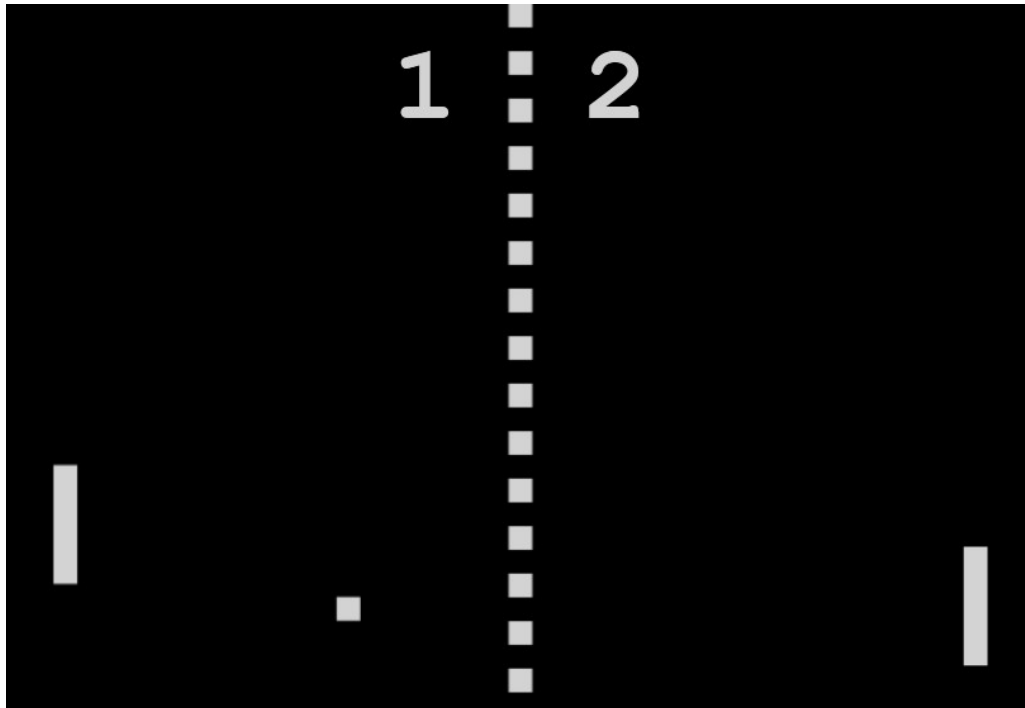
Detailed Game Specification: Pong

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1. General Game Overview

Pong is a competitive one or two-player tennis-like game that involves two paddles and a ball. The game is viewed from a top down perspective of the court, which is the entire screen. The game is tennis-like in that players rally the ball back-and-forth using their paddles, each trying to hit the ball past the other player's paddle. There is a centre line on the screen, but being a 2-D game the center line has no height and does not affect play in any way. There is a paddle on each vertical edge of the screen, each of which represents a player. In one-player mode, the controlling player is on the right side of the screen and the opponent is on the left side of the screen. In one-player mode the opponent is the computer, whereas in two-player mode the opponent is another human player. Each paddle is at a fixed horizontal position on the screen, but can be moved vertically up and down the full height of the screen. The objective of the game is to be the first to score three points. A point is scored when a player is unable to keep the ball in play, i.e., when a player is unable to deflect the ball back towards the opponent and the ball exits the screen on their edge. The current score is displayed in large digits near the top-center of the screen.



(from <https://www.ponggame.org/>)

The ball starts at the centre of the screen and is set in motion towards one of the players. The ball moves in a straight line, in two dimensions, at a set rate until it encounters a boundary or paddle. The boundaries are the four edges of the screen. Colliding with a top or bottom screen boundary or with a player's paddle causes the ball to rebound. Colliding with a left or right screen boundary results in a point being scored.

2. Game Play Details for Core 1-Player Version

Objectives and Rules

A game round starts with both player's paddles positioned with the centre of each paddle at the vertical centre of the screen. At the beginning of the first round, each player's score is zero, and the ball starts at the vertical centre of the centre line and immediately starts moving left with a horizontal velocity of 3 pixels per clock tick and a vertical velocity of 0.

The objective of the game is to be the first to score three points.

A point is scored when the opponent is unable to keep the ball in-play. This occurs when the leading edge of the ball (i.e., the left or right edge of the ball depending on the direction the ball is moving) contacts that vertical edge of the screen. For example, if the ball is moving left and the left edge of the ball contacts the left edge of the screen then a point is scored by the player on the right.

Ball Movement and Screen-edge Collisions

The ball is an object that is 8 pixels wide and 8 pixels high. The ball moves two dimensionally on the screen, meaning that there is both a horizontal and vertical velocity component to its motion. Velocity is measured in pixels per clock tick. The ball moves continuously in the given direction, unless it collides with an obstacle.

Colliding with either the top or bottom edge of the screen causes the ball to rebound vertically, such that the ball's horizontal velocity is unchanged, but the vertical velocity is reversed. Colliding with either the left or right edge of the screen will result in the end of the current point and the opponent being awarded a point. Colliding with the paddle will cause a rebound as described below.

In the event of a collision with either vertical edge of the screen play terminates. This results in the successful player's score being increased by one point. If that player's score reaches three then the game ends. If neither player's score is equal to three another round is started. This begins with both paddles in the same starting state as for round zero, as above, except the ball's initial direction alternates each time, regardless of which player scored the last point. As well, the ball starts in random position in the centre half of the screen and with a random horizontal and vertical velocity. Both of these will be values between 1 and 2.

When the game ends a message indicating whether the human player won or lost will be printed on the screen.

Paddle Movement and Collisions

The two paddles are 8 pixels wide and 48 pixels high. Both paddles are in fixed horizontal positions on the screen with 24 pixels between the back edge of the paddle and edge of the screen. Paddles can move vertically up or down. The right paddle is controlled by the human player and is moved using the up and down arrow keys. The player's paddle will move in the direction specified by the key that is currently depressed and will be stationary if no key is pressed. The left paddle is controlled by the computer. The computer paddle tracks the ball, meaning that it moves vertically with the ball, limited only by top and bottom screen boundaries. When in motion, each paddle moves at 8 pixels per clock tick.

A collision between the paddle and the top or bottom edge of the screen will stop the paddle, and the top or bottom edge of the paddle be left flush with the top or bottom edge of the screen (i.e., no portion of the paddle can move beyond the top or bottom edge of the screen).

The front edge of each paddle is divided into six equally sized segments, which are numbered from 1 to 6 with 1 being at the top of the paddle and 6 at the bottom. Collisions between the ball and the paddle have no effect on the paddle. However, such collisions will result in changes in the direction and/or the velocity of the ball. A collision between the ball and the front edge of the paddle will affect the ball depending on the segment in which the collision occurs. A collision in all of the segments will result in a reversal in the horizontal direction of the ball. A collision in segments 3 and 4 will result in both the horizontal and the vertical velocities being unchanged. A collision in either segment 2 or 5 will result in the horizontal velocity being set to 3 and the vertical velocity being set to 2. And collisions in either segment 1 or 6 will result in the horizontal velocity being set to 2 and the vertical velocity being set to 3.

The effect on the ball in a collision between the ball and the top or bottom edge of the paddle will depend on which half of the paddle the collision occurs. If the collision is in the front half of the paddle both the horizontal and vertical direction of the ball will reverse. If the collision is back half of the paddle only the vertical direction of the ball will reverse.

Objects

<u>Object or Object Type Name</u>	<u>Properties</u>	<u>Behaviours</u>	<u>Graphical Image</u>
Ball	• position (integer pair: x, y) • horizontal velocity (integer) • vertical velocity (integer) • horizontal direction (integer) • vertical direction (integer)	• move	[Small image.]

	<i>Note: both directions can be subsumed in the velocity properties if signed values are used.</i>	
Paddle	- size (constant integer pair: height, width) - position (constant integer: x, integer y) - vertical velocity (integer) - vertical direction (integer) <i>Note: the direction can be subsumed in the velocity property if signed values are used.</i> - size (constant integer pair: height, width)	move
Score	- position (constant integer pair: fixed x, y) - size (constant integer pair: height, width) - value (integer)	update

Asynchronous (Input) Events

<u>Event Name</u>	<u>Triggering Input Event</u>	<u>Description</u>
Move up request	Up arrow key is depressed	This will set the player's paddle vertical direction to -1.
Move down request	Down arrow key is depressed	This will set the player's paddle vertical direction to 1.

Quit	Esc key is depressed	Will set the terminate game flag
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Synchronous (Timed) Events

<u>Event Name</u>	<u>Trigger Timing</u>	<u>Description</u>
Move ball	Every 1/70 th of a second.	The ball moves horizontally <i>horizontal velocity*horizontal direction</i> pixels and vertically <i>vertical velocity*vertical direction</i> pixels. This will update the x,y position properties in the ball object.
Move player paddle	Every 1/70 th of a second.	The player's paddle y position property will be incremented vertically by <i>vertical velocity*vertical direction</i> pixels. After updating the player's paddle position properties the player's paddle vertical direction property will be set to zero.
Move computer paddle	Every 1/70 th of a second.	The vertical distance between the computer paddle position and the ball position will be calculated. Based on whether the vertical distance is above, the same as or below the paddle the computer's paddle vertical direction will be set to either -1, 0 or 1. The computer's paddle y position property will be incremented vertically by <i>vertical velocity*vertical direction</i> pixels. After updating the computer's paddle position properties the computer's paddle vertical direction property will be set to zero.

Condition-Based (Cascaded) Events

<u>Event Name</u>	<u>Triggering Condition</u>	<u>Description</u>
Ball – top edge collision	The ball has moved and the ball's top edge is less than or equal to the screen's top edge (0).	The ball's y position is reset so that the ball's top edge is flush with the screen's top edge and the ball's vertical velocity is reversed, i.e. $\text{vertical_velocity} * -1$
Ball – bottom edge collision	The ball has moved and the ball's bottom edge is greater than or equal to the screen's bottom	The ball's y position is reset so that the ball's bottom edge is flush with the screen's bottom edge and the ball's vertical velocity is reversed, i.e. $\text{vertical_velocity} * -1$

	edge (399).	
Ball – left edge collision	The ball has moved and the ball's left edge is less than or equal to the screen's left edge (0).	The player's score is incremented by one. Play is terminated and a check is done to determine whether the game is over or resetting to start a new round is required.
Ball – right edge collision	The ball has moved and the ball's right edge is greater than or equal to the screen's right edge value (639).	The player's score is incremented by one. Play is terminated and a check is done to determine whether the game is over or resetting to start a new round is required.
Win or reset	The ball has collided with the right or left edge of the screen.	If the scoring player's score is equal to three the terminate game flag is set to true and a message indicating whether or not the player won is displayed. Otherwise the position of both paddle are reset to the starting locations and the ball's position and velocity are reset using randomly generated values. After setting the ball velocity the starting direction is alternated, i.e. $\text{direction} * -1$
Paddle –top edge collision	The paddle has moved and the paddle's top edge is less than or equal to the screen's top edge (0).	The paddle's y position is reset so the paddle's top edge is flush with the screen's top edge.
Paddle –bottom edge collision	The paddle has moved and the paddle's bottom edge is greater than or equal to the screen's bottom edge (399).	The paddle's y position is reset so the paddle's bottom edge is flush with the screen's bottom edge.
Ball –paddle top edge collision	Either the ball or the paddle has moved and the bottom edge of the ball is less than or equal to the top edge of the paddle.	For the player's paddle: If the bottom-right corner of the ball is less than the middle of the paddle width then both the horizontal and vertical velocities of the ball are reversed, i.e. both velocities $* -1$. Otherwise, if the ball's bottom-right corner is greater than or equal to the middle of the paddle then only the vertical velocity is reversed, i.e. $\text{Vertically_velocity} * -1$

		For the opponent's paddle: The exact same behavior is seen except it is if the ball's bottom-left corner is greater than the middle of the paddle width for a both reversal or the ball's left-bottom corner is less than or equal to the middle of the paddle for only a vertical reversal.
Ball –paddle bottom edge collision	Either the ball or the paddle has moved and the top edge of the ball is greater than or equal to the bottom edge of the paddle.	For the player's paddle: If the top-right corner of the ball is less than the middle of the paddle width then both the horizontal and vertical velocities of the ball are reversed, i.e. both velocities * -1. Otherwise, if the ball's top-right corner is less than or equal to the middle of the paddle then only the vertical velocity is reversed, ie. vertically_velocity * -1
Ball –paddle front edge collision	Either the ball or the paddle has moved and either side edge of the ball in the direction the ball is moving is past or equal to the front edge of the paddle.	The side edge of the ball in the direction the ball is moving is the right edge if the ball is moving right or the left edge if the ball is moving left. The height of each paddle is 48 pixels. The paddle is divided into 6 equal height segments, so each segment is 8 pixels high. Pixels are numbered from 0 to 47 with the numbering starting from the top of the paddle. The segments are numbered 1 to 6, also starting from the top of the paddle. In all cases a collision of the ball with the front edge of the paddle results in a change in a rebound. Depending on which segment the ball collides with determines the type of rebound: – a collision in segments 3 and 4: result in a rebound where the vertical velocity changes to 0 and horizontal velocity reverses direction, i.e. horizontal_velocity * -1 – a collision in segments 2 and 5 result in a rebound that reverses both the horizontal and vertical directions and changes the horizontal velocity to 3 and the vertical velocity to 2 – a collision in segments 1 and 6 result in a rebound that reverses both the horizontal and vertical

directions and changes the horizontal velocity to 2 and the vertical velocity to 3
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Hypothetical Gaming Session

The game begins with both paddles stationary, slightly out from their respective edge of the screen and centred vertically on the screen. The score for each player, 0 respectively is displayed just offset from the centre line on each player's side. The ball appears at the centre of the screen moving slowly horizontally left with no vertical movement. The player does not press either up or down arrow keys and the ball strikes the centre of their paddle and rebounds straight back. The opponent's paddle moves slightly down so that the ball hits towards the end the paddle and the ball rebounds going upward and to the right with a vertical velocity of 2 and horizontal velocity of 3. The player presses the up repeatedly moving their paddle up tracking the ball. The ball collides the the top edge close to the centre line and rebounds still going right but not going down and with the same horizontal and vertical velocities. The player is slow to react to the change in vertical direction and by the time they start pressing the down arrow to move the paddle downward the ball reaches the left edge of the screen and the opponent scores a point. The screen changes to the same starting setup but the ball appears a third of the way from the bottom of the screen and moving to the left and upward. The opponent's paddle initially moves down due to the location of the ball and as the ball moves upward the opponent's paddle changes to going up, but is too slow and the ball collides with the left edge of the screen and the player scores a point. Again the screen changes to the starting setup and this time the ball appears a quarter of the way from the top of the screen and moving to the right and downward. The player presses the down arrow key repeatedly moving the paddle so the bottom of the paddle is flush with the bottom edge of the screen. Once at the bottom of the screen repeated presses of the down arrow key have no effect on the paddle, it is unable to move below the bottom edge of the screen. The ball collides with the player's paddle just before the bottom of the paddle, which results in a rebound where the ball's motion changes to going left and still down, but with a horizontal velocity of 2 and vertical velocity of 3. The ball almost immediately collides with the bottom edge of the screen, which results in a rebound where the horizontal direction is still left and velocity values are unchanged but the vertical direction is now upward. The opponent's paddle has been tracking the ball and is still moving down. Once the ball reaches a point about a third of the way from the bottom of the screen the opponent's paddle starts moving upward. Just after the centre line the ball collides with the top edge of the screen. Again the ball's horizontal direction remains going left and with the same velocities, but the vertical direction changes to down. The opponent's paddle continues to move upward, but then changes and goes down and has the ball collide with it about a third of the way down the paddle. This results in the ball's vertical direction remaining going down, but the horizontal direction changes to going right and the horizontal velocity changes to 3 and the vertical direction changes to 2. Both players are able to move their paddles to collide with the ball sending it back and forth with various angles. This continues until the opponent is unable to move to block the ball and it collides with the left edge of the screen. This results in the player's score incrementing to 2 and the game resetting. Another round continues until again the opponent is unable to keep the ball in play and the player's score is incremented to 3. This results in the screen remaining the same, but a message is printed in the middle of the screen congratulating the player on winning.

The player feeling good about themselves decides to play again. Paly resumes with the same back and forth movement of the ball, but in this case the player is unable to score as frequently and the opponent is able to score three points first. In the case the game ends a message telling the player better luck next time.

3. Game Play Details for Core 2-Player Version

In the two player game both players are human and on separate computers. All of the rules remain the same with the only difference is in both cases the opponent , i.e. the player on the left side of the screen, is a human controlling the paddle.

[2-player sample screenshot goes here.]

4. Sound Effects

<u>Sound Effect Name</u>	<u>Brief Description</u>	<u>Event which Triggers Playback</u>
The ball rebounding of an object	boing	ball-paddle collision Ball – top or bottom edge collision
The ball going out of play	Ding	The ball colliding with a side edge
Game over	Dun-dun-dun	Win or reset with game over being true

The background music be a simple repetition of three notes

5. Additional Features (Time Permitting)

Have the velocity be maintained and build upon each other to a maximum velocity in each direction.

Increased segments with different rebound characteristics.

Maintain a counter such that after some number of consecutive hits with the same segment pair the paddle size is increased.